INCH-POUND

MIL-T-11838E
5 October 1993
SUPERSEDING
MIL-T-11838D
13 September 1972

MILITARY SPECIFICATION

TREADWAY, SUPERSTRUCTURE: BRIDGE, FLOATING; FOOT, ALUMINUM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the treadway of the floating aluminum footbridge.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

TT-E-527 - Enamel, Alkyd, Lusterless.

MILITARY

MIL-T-704 - Treatment and Painting of Materiel.
MIL-W-5044 - Walkway Compound, Nonslip, and Walkway Matting, Nonslip.

STANDARDS

FEDERAL

FED-STD-595 - Colors Used in Government Procurement.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-889 - Dissimilar Metals.

Beneficial comments (recommendations, additions, deletions) and any pertintent data which may be of use in improving this document should be addressed to: US ARMY BELVOIR RDE CTR, ATTN SATBE TSE, 10101 GRIDLEY RD STE 104, FT BELVOIR VA 22060-5818 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5420

AMSC N/A
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(Unless otherwise indicated, copies of federal and military specifications and standards are available from: STDZN DCMNT ORDER DESK, BLDG 4D, 700 ROBBINS AVE, PHILADELPHIA PA 19111-5094.)

2.1.2 Other Government drawings. The following Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

DRAWINGS

ME

TA13211E8560 - Treadway, Superstructure: Bridge Floating.
13207E9958 - Treadway, Superstructure: Bridge Floating, Foot, Aluminum, Packing of.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from: USA BELVOIR RDE CTR, ATTN SATBE JBS, 10101 GRIDLEY RD, STE 104, FT BELVOIR VA 22060-5818.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN WELDING SOCIETY (AWS)

A5.10 - Bare Aluminum and Aluminum Alloy Welding Electrodes and Rods.
D1.2 - Structural Welding Code, Aluminum.

(Application for copies should be addressed to: AMERCN WELD SCTY, 550 NW LEJEUNE RD, PO BOX 351040, MIAMI FL 33135.)

UNIFORM CLASSIFICATION COMMITTEE (UCC)

Uniform Freight Classification Rules.

(Application for copies should be addressed to: UNFRM FRGHT CLASS CMMTE, ATTN TARIFF PBLSHNG OFCR, ROOM 1106, 222 S RIVERSIDE PLZ, CHICAGO IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT (NMFTA)

National Motor Freight Classification Rules.

(Application for copies should be addressed to: NATL MTR FRGHT TRFC ASS, ATTN TRFC ORDR SECT, 2200 MILL RD, ALEXANDRIA VA 22314.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The treadway shall be as shown on TA13211E8560 and as specified (see 6.2).

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviations from the prescribed dimensions or tolerances is

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permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly and operation of the treadway. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or the designated representative.

3.2 First article. Unless otherwise specified (see 6.2) a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials. Materials shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The treadway shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the treadway may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the treadway may be newly fabricated from recovered materials to the maximum extent practicable, provided the treadway produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the treadway.

3.4 Identification marking. The treadway shall be identified in accordance with MIL-STD-130 and shall be metal-die stamped as shown on the applicable drawing. Die stamp shall be legible after painting.

3.5 Treatment and painting. All parts of the treadway shall be cleaned, treated, and painted as specified on the drawings except that the color number designation shall be 34088 in accordance with FED-STD-595. Contact surfaces, regardless of whether such contact is between similar or dissimilar metals, shall be cleaned, treated, and painted prior to assembly. The raised portions of the entire top of the corrugated deck shall be surfaced, not less than 72 hours after painting, with one coat of nonslip material conforming to MIL-W-5044, type II, color number 34088. Following painting and surfacing, white paint (color number designation 37925) conforming to TT-E-527 shall be applied as follows:

a. A stripe running the length of the corrugated deck on the centerline, 3 inches wide, which may be applied to the raised portions only.
b. The exposed upper surface of the upper handrail brackets.
c. The machined surfaces in the throat of female connector plates from the end to a point 2 inches from the end.
d. The inner exposed web surfaces of the girders (find numbers 1 and 2) outlined by the end of the girder and the vertical centerline of the retainer lug.

3.6 Stenciling. The nomenclature, National Federal Stock Number, weight, and cube shall be stenciled on each treadway. The letters "NET WT" shall precede the numerals in the weight and the letters "CU" shall precede the cubic displacement. The markings shall be Gothic capitals and Arabic numerals, 0.750-inch high. Stenciling shall conform to MIL-T-704.

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3.7 Workmanship. The treadway or components shall be free from any irregularities which would interfere with its operation.

3.7.1 Aluminum fabrication. Before any work or layout is done, aluminum shall be straightened by methods that will not result in injury to the metal. Care shall be used in handling aluminum during fabrication to avoid excessive scratching or damage to edges. Finished members shall be free from laminations, kinks, bends, scratches, or winds. Corners shall be square and true. Aluminum 0.375-inch thick and over shall be cut by routing or sawing. Aluminum under 0.375-inch thick may be cut by shearing. Aluminum shall not be cut by methods using heat.

3.7.2 Riveted connections. A jig frame shall be employed in making riveted connections to insure alinement and maintain required tolerances. Assemblies bolted prior to riveting shall show no undue separation. Rivets shall be driven cold with pressure tools, where practicable, or with a pneumatic hammer. The driven head, except as otherwise specified on the drawings, shall be button or cone point when formed with a squeeze riveter or pneumatic hammer or squash or flathead when formed by a bucking bar or dolly. The countersunk heads for locator plates may be driven or manufactured. Rivet heads shall be full, concentric with the rivet shank axis, and shall have full bearing on the surface of the member.

3.7.3 Welding. The surfaces of parts to be welded shall be free from scale, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subject to proof and service loadings. Welds shall be free from such defects as cracks, flux inclusions, spatter, undercutting, porosity, lack of fusion with parent metal, or incorrect size.

3.7.3.1 Welders or welding operators. Welders and welding operators assigned to manual welding work covered by this specification shall be qualified in accordance with AWS D1.1 Structural Welding Code Aluminum, for the materials joined and the type of welding operation to be performed. The contractor is responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS codes (see 6.4).

3.7.3.2 Welding electrode. Welding electrodes for aluminum shall conform to AWS A5.10, classification ER4043, size as required.

3.7.4 Interchangeability. Treadways shall be fabricated to be duplicates within the tolerances specified and shall be interchangeable.

3.8 Government-loaned property. Unless otherwise specified (see 6.2) the following property in the quantities indicated in table I will be loaned by the Government (see 6.7).

TABLE I. Government-loaned property.

Item	Description	Identification	Quantity
1.	Inspection equipment for treadway	Quality Assurance Pamphlet, P715-108	1 set
2.	QA Pamphlet, care, operation, and maintenance of inspection equipment for treadway, superstructure	P715-108	1

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensureing that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.5 and 6.6.1).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article examination. The first article treadway shall be examined as specified in 4.4.2.1, as applicable and for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.2 Examination.

4.4.2.1 Individual. Each treadway shall be inspected for conformance to the dimensions covered by the quality assurance provisions (QAP) shown on the drawings under "Major - 100 percent inspection". Unless otherwise specified (see 6.2), Government-loaned property (see 3.8), shall be used in accordance with Government-loaned QA pamphlet (see 3.8), to inspect dimensions specified on the applicable drawings. Nonconformance to any of these dimensions shall constitute failure of this examination and shall be cause for rejection.

4.4.2.2 Samples. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1.

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4.5 Inspection procedure.

4.5.1 Examination. The treadway shall be examined as specified herein for the following defects.

Major

101. Material not as specified (see 3.3).
102. Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment as specified (see 3.3.1).
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other as specified (see 3.3.1.1).
104. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the treadway (see 3.3.2).
105. All dimensions designated major on the QAP other than those under 100 percent inspection not as specified (see 4.1.3).
106. Components missing or not as specified (see 4.1.2).
107. Workmanship not as specified (see 3.7).
108. Riveting not as specified (see 3.7.2).
109. Weld defects such as flux inclusions, spatter, undercutting, porosity, incorrect size, or lack of fusion with parent metal (see 3.7.3).
110. Nonslip covering not applied to corrugated panels as specified (3.5).
111. Welders or welding operators not qualified as specified (see 3.7.3.1).
112. Welding electrode not as specified (see 3.7.3.2).

Minor

201. All dimensions not designated major on the QAP not as specified (see 4.1.3).
202. Identification marking incorrect, missing or illegible (see 3.4).
203. Treatment and painting not as specified (see 3.5).
204. Stenciling not as specified (see 3.6).
205. Bolts not peened or staked as specified on drawings (see 3.1.1).

4.6 Inspection of packaging. The packing and marking shall be examined to determine compliance with 13207E9958 and section 5 of this specification. Any packing that is not in accordance with the drawing or any marking that is missing, illegible, incorrect, or incomplete shall constitute failure and shall be cause for rejection.

5. PACKAGING

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Levels A and B. The treadways shall be packed in accordance with 13207E9958.

5.1.2 Level C. The treadways shall be packed to assure carrier acceptance and safe delivery to destination at lowest rates in compliance with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.2 Marking. In addition to any special marking specified in the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The treadway is intended for use as the deck section of the floating aluminum footbridge.

6.2 Acquisition requirements. Acquisition documents should specify the following:

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a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When first article is not required for inspection and approval, and the number of units required (see 3.2).
d. Level of packing required (see 5.1).
e. When Government-loaned property is not required (see 3.8).
f. Any special marking (see 5.2).

6.3 First article. When a first article inspection is required, the item(s) should be a production model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD Far Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title
3.7.3.1	DI-MISC-80876	Welding Procedure Qualification Test Report
3.7.3.1	DI-MISC-80875	Welding Procedures

The above DIDs were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.5 Quality assurance provisions (QAP). The contracting officer should require the supplier to maintain record of all QAP inspections. A suggested paragraph is as follows:

"The supplier shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, corrected action taken when applicable."

6.6 Definitions.

6.6.1 Quality Assurance Provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.7 International standardization. Certain provisions of this specification are the subject of International Standardization Agreement ABC, Standard 88. When amendment, revision or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels, including

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departmental standardization offices, to change the agreement or make other appropriate accommodations.

6.8 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.8.

6.9 Welders and welding operators. The contracting officer should consider requalifying welders or welding operators in the event of evidence of poor welds (see 3.7.3)

6.10 Subject term (key word) listing.

Footbridge, aluminum
Footbridge, floating

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

User activity:
Navy - MC

Preparing activity:
Army - ME

Project 5420-0170

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
   The submitter of this form must complete blocks 4, 5, 6, and 7.
   The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-T-11838E	2. DOCUMENT DATE (YYMMDD) 931005

3. DOCUMENT TITLE
Treadway, Superstructure, Bridge, Floating, Foot, Aluminum

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) DSN	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Carolyn B. Johnson	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3468 (2) DSN 654-3468
c. ADDRESS (Include Zip Code) ARMY BELVOIR RDE CTR ATTN SATBE TSE 10101 GRIDLEY RD STE 104 FT BELVOIR VA 22060-5818	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: DEFNS QLTY & STDZN OFC 5203 LEESBURG PIKE STE 1403 FLS CHURCH VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

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